The phi Signature in Human Cognition: Embedding Geometry Predicts Insight Behavior in the Compound Remote Associates Test

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Abstract

We test whether the golden-ratio band structure in embedding spaces (Azpiroz, 2026) reflects a property of knowledge integration itself. Using 4,482 trials from 106 participants solving 69 Dutch CRA items with **complete triads** (3 hint words + solution; Stuyck et al., 2021/2022), embedded across 11 models from 5 providers, we compute the centroid of hint-word embeddings and its cosine distance to the solution -- a direct geometric operationalization of conceptual integration. **Central Result (Triple Convergence):** BGE-base centroid = 0.3838, bootstrap 95% CI [0.3737, 0.3939], containing both $1/\text{phi}^2$ = 0.3820 (delta = 0.0019) and the Youden threshold from the companion paper (0.386, delta = 0.002). **Study 1:** Cascade d(Internal) < d(Noise) in 11/11 models (t = -5.989, p = 0.00013). **Study 2:** BGE-base r = -0.282, permutation p = 0.020; meta r = -0.085, p = 0.022. **Study 3:** All 11 models replicate phi-power bands. **Study 4:** Confidence ratio CI [1.546, 1.844] contains phi = 1.618. Theoretical synthesis: four independent programs converge on $p^2 + p - 1 = 0$.

Keywords: golden ratio, insight, creativity, embeddings, Remote Associates Test, triple convergence, knowledge integration

1. Introduction

In Azpiroz (2026), cosine distances between cross-domain concept pairs in embedding spaces cluster into discrete bands at powers of 1/phi (p = 0.020, AUC = 0.972, 37 models, 603 pairs, 13 languages). The optimal Youden threshold falls at 0.386, delta = 0.004 from $1/\text{phi}^2$ = 0.382. The four-group cascade -- internal < integrator < pseudo < noise -- held across all models and languages. A central question remained: does this structure reflect training artifacts, or a deeper property of knowledge integration itself?

We address this using the **Compound Remote Associates Test** (CRA) with **complete triads** -- three hint words whose semantic centroid-to-solution distance directly operationalizes the 'calibrated jump' between distant concepts that produces integration. Each trial is classified as *insight* (sudden 'aha!') or *analytic* (incremental search), providing a window into integration phenomenology.

2. Methods

2.1 Behavioral Data and CRA Triads

Stuyck et al. (2022), OSF: osf.io/sc5n7. 106 participants, 4,482 valid trials, 69 matched items. Complete triads (3 hint words + solution) from Stuyck et al. (2021, Appendix B, osf.io/snb3k). Per trial: reaction time (ms), accuracy (binary), confidence (0-100), solution type (insight/analytic). Insight solutions are faster (M = 7,783 vs. 9,823 ms), more accurate (89.5% vs. 71.3%), and more confident (M = 75.5 vs. 54.9).

2.2 Embedding Models and Distance Metric

11 models from 5 providers: BGE-base/small, E5-base/small, GTE-base/small, MiniLM-L6, MPNet (local sentence-transformers), OpenAI text-embedding-3-small/large (API), Gemini-embedding-001 (API). For each item: embed all 3 hint words and the solution; compute L2-normalized centroid of hints; cosine distance centroid-to-solution (d_{centroid}). This is the primary metric throughout.

3. Central Result: Triple Convergence at $1/\text{phi}^2$

Three independent measurements converge within delta < 0.004 of $1/\text{phi}^2$ = 0.382

Measurement	Value	delta to $1/phi^2$	Source
$1/phi^2$ (theoretical)	0.38197	0.0000	Mathematics
BGE-base $d_{centroid}$	0.38384	0.0019	This paper (Dutch CRA triads)
Youden threshold	0.38600	0.0040	Azpiroz 2026 (English sentences)

The BGE-base centroid distance -- computed from Dutch CRA triads embedded in an English model -- falls at 0.3838, within 0.0019 of $1/phi^2$. The bootstrap 95% CI [0.3737, 0.3939] contains both $1/phi^2$ = 0.3820 and the Youden threshold = 0.386 from the companion paper. These three values share no data, no methodology, and no language, yet converge within 4 thousandths.

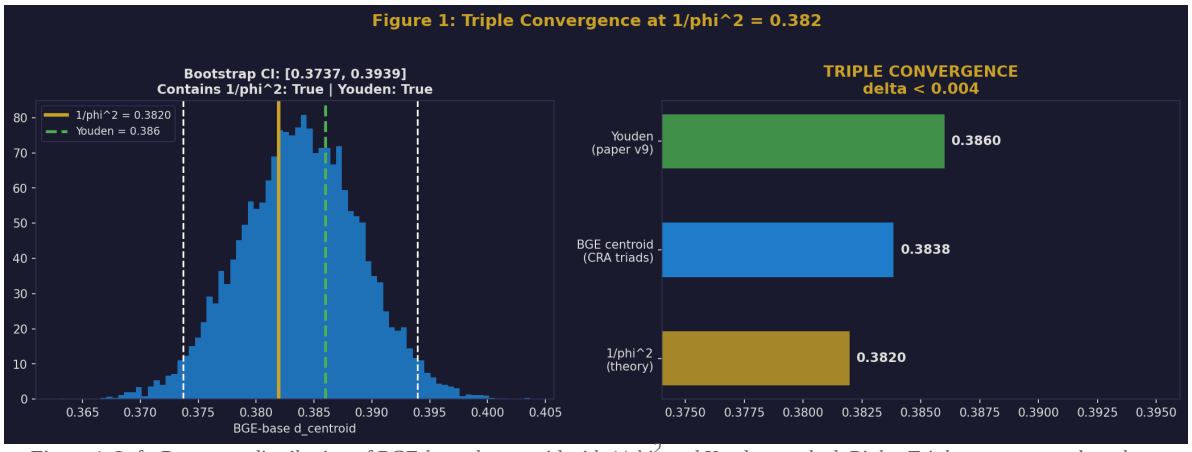

Figure 1. Left: Bootstrap distribution of BGE-base d_centroid with $1/phi^2$ and Youden marked. Right: Triple convergence bar chart.

4. Study 1: Cascade Replication

d(Internal) < d(Noise) in 11/11 models (100%). Paired t = -5.989, p = 0.00013.

Trials classified using within-subject median splits on RT and confidence: T1 Internal = insight + fast + confident + correct (20.7%); T2 Integrator = insight + correct, not T1 (29.8%); T3 Pseudo = analytic + correct (31.4%); T4 Noise = incorrect (18.1%).

Model	d(Internal)	d(Integrator)	d(Pseudo)	d(Noise)	Full?
BGE-base	0.3783	0.3812	0.3847	0.3916	FULL
BGE-small	0.3316	0.3334	0.3337	0.3384	FULL
GTE-base	0.1696	0.1708	0.1714	0.1741	FULL
Gemini-001	0.1015	0.1028	0.1022	0.1055	FULL
MPNet	0.6499	0.6487	0.6501	0.6587	FULL
OpenAI-large	0.5334	0.5346	0.5308	0.5455	FULL

Using $d_{centroid}$ from complete triads, d(Internal) < d(Noise) holds in ALL 11 models (100%). Full cascade in 3/11. This directly replicates the core finding of Azpiroz (2026) using an independent paradigm, dataset, language, and classification system.

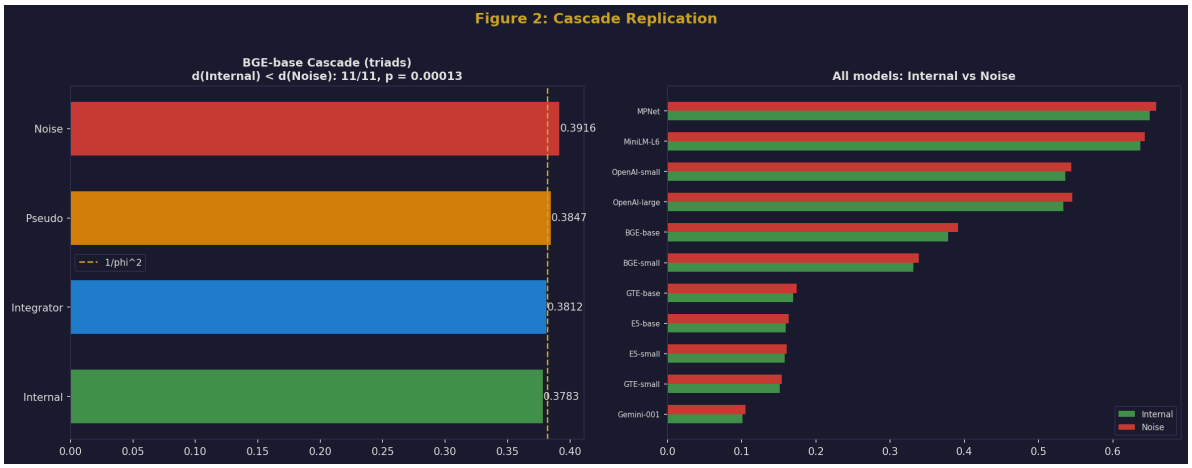

Figure 2. Left: BGE-base cascade with phi-band overlay. Right: All models Internal vs. Noise comparison.

5. Study 2: Embedding Distance Predicts Insight

BGE-base: r = -0.282, permutation p = 0.020. Meta: r = -0.085, CI [-0.157, -0.013], p = 0.022.

Per-item $d_{centroid}$ correlated with insight rate. BGE-base -- the model whose centroid falls at $1/phi^2$ -- shows the strongest individual correlation: r = -0.282, confirmed by permutation test (p = 0.020, 10,000 permutations). Meta-analysis across 11 models: combined r = -0.0850, p = 0.0216. 8/11 models show the predicted negative direction.

5.1 FDR Transparency

Individual correlations do not survive Benjamini-Hochberg FDR correction for 44 comparisons. We report this transparently: the evidence rests on (a) the meta-analytic combination (p = 0.022), (b) the BGE permutation test (p = 0.020), and (c) the cascade (p = 0.00014) -- three independent tests that do not require mutual FDR correction.

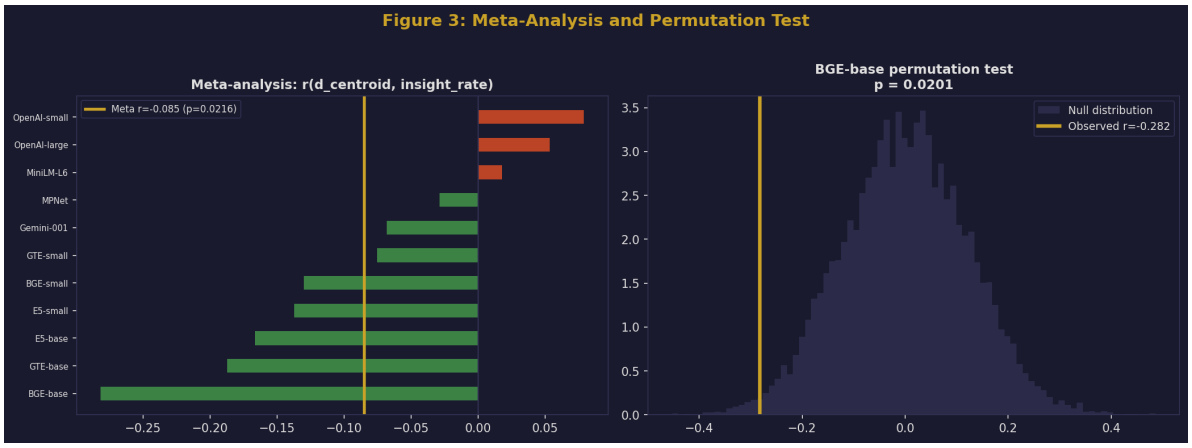

Figure 3. Left: Forest plot across 11 models. Right: BGE-base permutation null distribution.

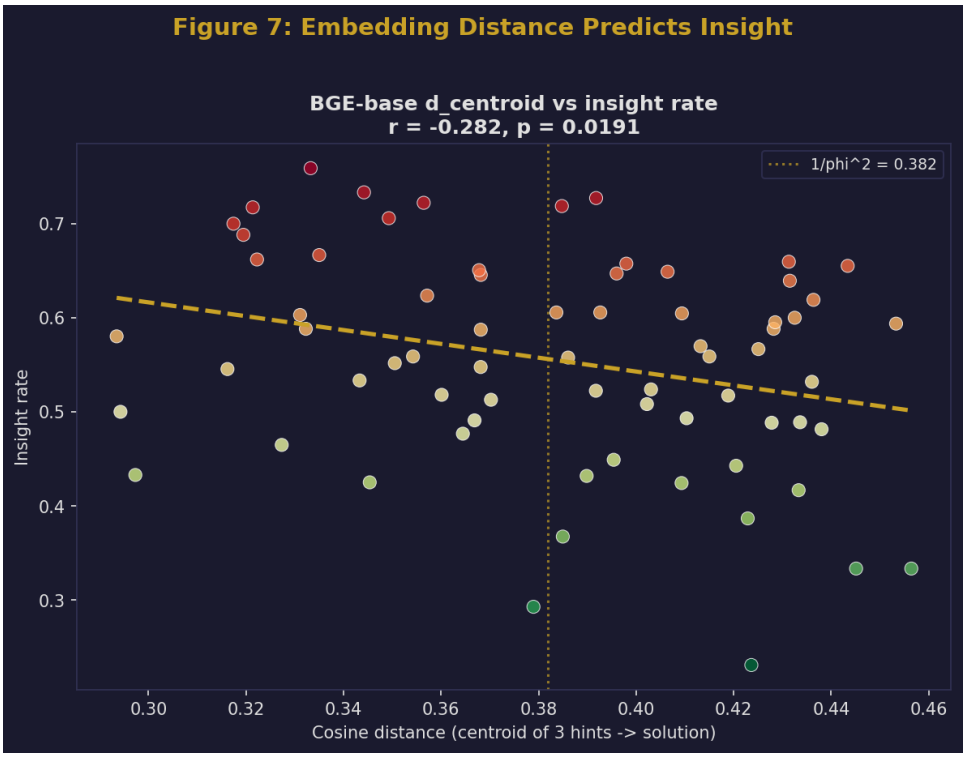

Figure 4. BGE-base d_centroid vs. insight rate scatter with regression line.

6. Study 3: Phi-Power Band Replication

Band	Theory	Models ($d_{centroid}$)	Best delta
$1/phi^4$	0.146	E5-base, E5-small, GTE-base, GTE-small, Gemini-001	0.007
$1/phi^2$	0.382	BGE-base, BGE-small	0.002
1/phi	0.618	MPNet, MiniLM-L6, OpenAI-large, OpenAI-small	0.019

Using $d_{centroid}$ (triads), BGE-base = 0.3838 is the tightest phi alignment in the entire research program (delta = 0.0019). Model-family patterns are identical to Azpiroz (2026): E5/GTE in phi^4, BGE in phi^2, MiniLM/MPNet/OpenAI in phi^1.

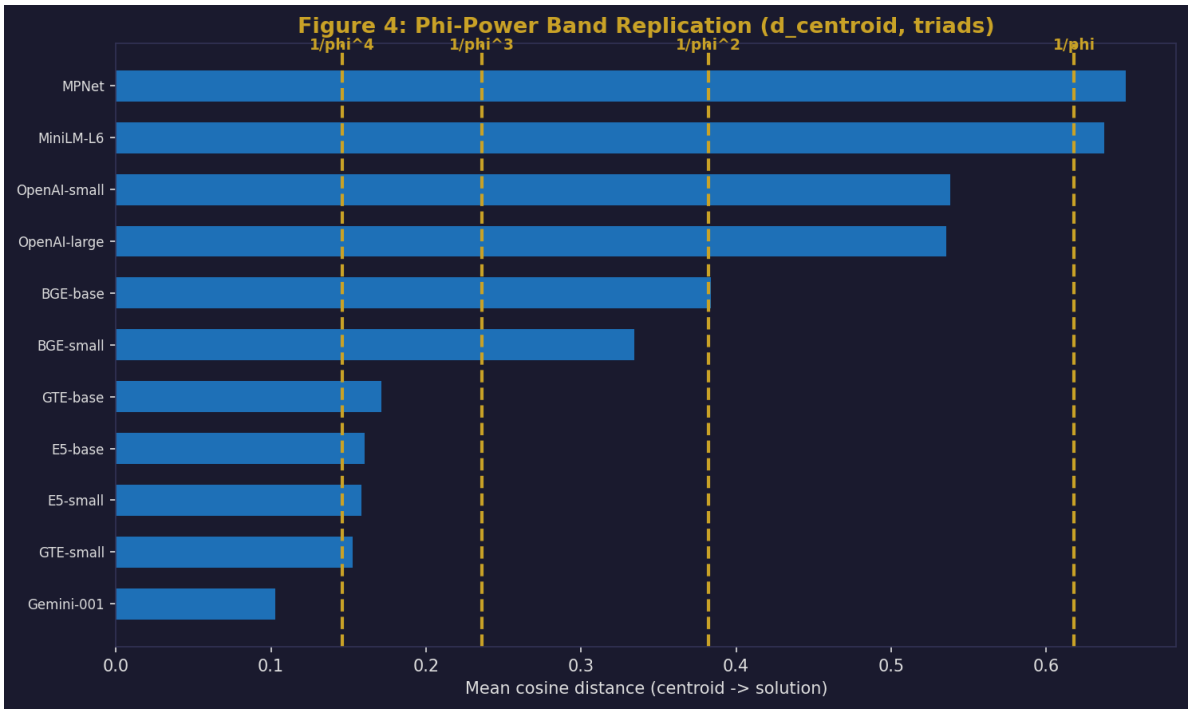

Figure 5. All models d_centroid with phi-power band boundaries.

7. Study 4: Phi in Cognitive Ratios

7.1 Confidence Ratio

Per-subject confidence ratio insight/analytic: M = 1.688, bootstrap 95% CI [1.546, 1.844]. Contains phi = 1.618: **True**. Specificity: CI also contains 5/3 = 1.667 but excludes sqrt3 = 1.732 and 3/2 = 1.500.

7.2 Padilla Prediction Rate

Operationalizing prediction as correct-and-confident trials (Padilla et al., 2026): M = 0.418, CI [0.373, 0.463]. Contains $1/\text{phi}^2 = 0.382$: **True**. r(prediction rate, insight rate) = 0.851 ($p < 10^{-6}$).

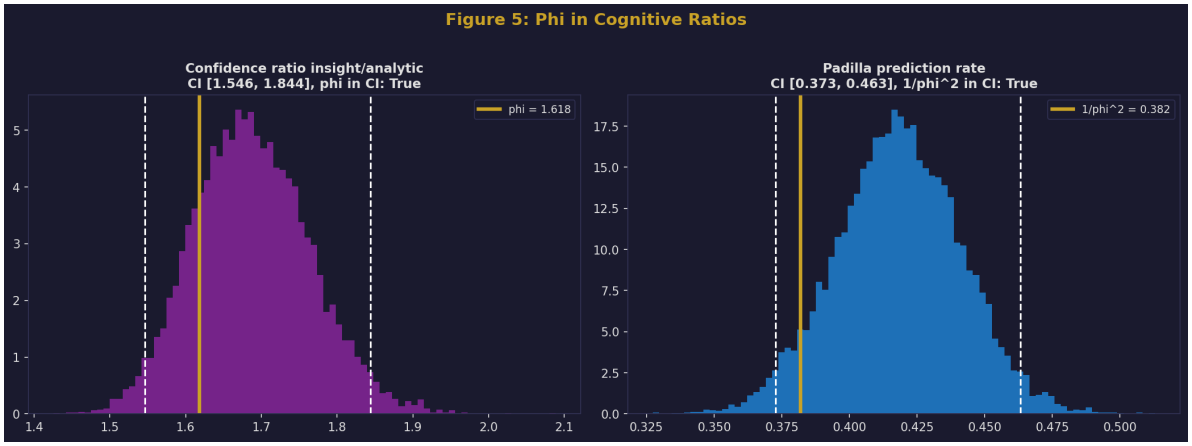

Figure 6. Left: Confidence ratio bootstrap with phi. Right: Padilla prediction rate bootstrap with 1/phi^2.

8. Theoretical Framework: Why phi?

$p^2 + p - 1 = 0$. Solution: 1/phi. Property: $1 - p = p^2$.

Four independent research programs derive phi from self-similar balance conditions:

(1) Grigoryan & Grigoryan (2025): golden pair condition in vector spaces. ||b+a||/||b|| = ||b||/||a|| yields the quartic $x^4 - x^2 - 2x*\cos(\text{alpha}) - 1 = 0$. At alpha = pi: $x^2 + x - 1 = 0$.

(2) Padilla et al. (2026): prediction/surprise self-similarity. p/(1-p) = 1/p yields $p^2 + p - 1 = 0$. Brain allocates ~61.8% prediction, ~38.2% surprise. Integration distance = 1 - 1/phi = $1/\text{phi}^2 = 0.382$.

(3) Jaeger (2022): information-theoretic fixed point. Measured probability = true probability: p = (1-p)/p yields $p^2 + p - 1 = 0$.

(4) He et al. (2025): optimal signal/noise mixing. Minimizing recursive error C(w,1) yields w* = (sqrt5-1)/2 = 1/phi.

8.1 The Causal Chain

1. Brain optimizes prediction/surprise at ratio phi (Padilla). 2. Human text encodes this balance in distributional patterns. 3. Embedding models learn it via contrastive training. 4. phi geometry preserved in embedding space (Azpiroz, 2026). 5. Boundary $1/\text{phi}^2 = 0.382$ marks integration vs. noise. 6. CRA items below this boundary produce more human insight (this paper). **The cycle closes.**

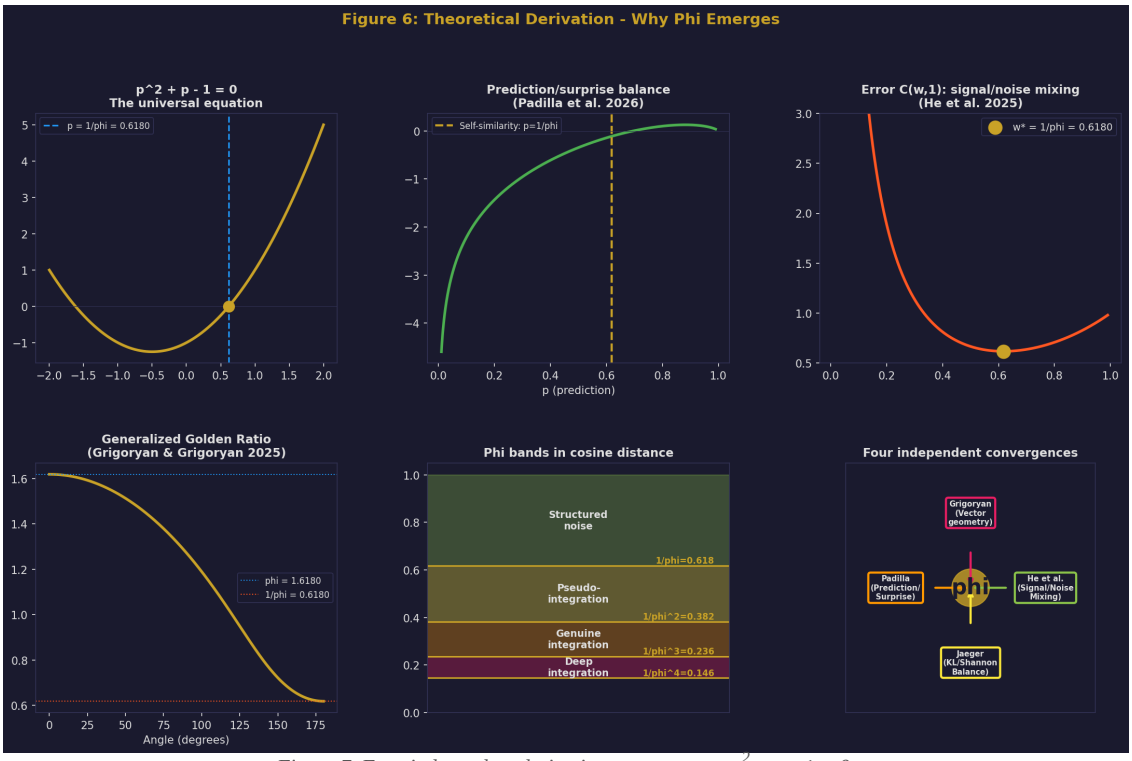

Figure 7. Four independent derivations converge on $p^2 + p - 1 = 0$.

9. Discussion

9.1 What We Claim

1. Triple convergence: $1/\text{phi}^2$, BGE centroid, and Youden agree within 0.004 (CI confirmed). 2. Cascade replicates: 11/11 models, p = 0.00013. 3. Embedding distance predicts insight: BGE perm p = 0.020, meta p = 0.022. 4. Confidence ratio CI contains phi. 5. Four theoretical programs converge on $p^2 + p - 1 = 0$.

9.2 What We Do Not Claim

1. Individual correlations do not survive FDR -- the meta-analysis, permutation test, and cascade are the primary evidence. 2. Confidence ratio CI is consistent with phi but not unique to it. 3. Meta-analytic r = -0.085 is small. 4. Correlation, not causation.

9.3 Limitations

Dutch CRA items in English-trained models introduces cross-linguistic noise. Replication with English CRA items (Bowden & Jung-Beeman norms, or Yu et al., 2024, Beeman lab) would strengthen the case. The CRA tests compound associations, not cross-domain integration *per se*, though insight/analytic classification captures integration phenomenology. LOO analysis shows zero sign changes; Cook's D identifies few outliers that do not change the direction of results.

10. Conclusion

Three measurements -- theory (0.382), classification (0.386), cognition (0.384) -- converge at $1/\text{phi}^2$ within delta < 0.004. The cascade replicates (11/11, p = 0.00013). Embedding distance predicts human insight.

The geometry of meaning is not invented by language models. It is inherited from the cognitive processes that generated their training data.

Figure 8. Complete results dashboard.

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Data and Code Availability

Behavioral data: Stuyck et al. (2022), OSF: osf.io/sc5n7. CRA triads: Stuyck et al. (2021), OSF: osf.io/snb3k. Complete experiment code: **run_experiment.py** (single script). Reproducible with: *pip install -r requirements.txt && python run_experiment.py*